

Retail Sales & Inventory Intelligence System – Project Report

1. Datasets Used -

The project is based on a retail company selling mobile phones and laptops. The dataset contains nine CSV files: **Orders, Order_Items, Customers, Stores, Staffs, Products, Stocks, Brands, and Categories.**

Each dataset captures essential business operations such as sales transactions, product details, stock levels, store information, and staff performance. The data is structured for relational database use and analytics.

2. Excel – Data Preprocessing -

The raw data from multiple CSV files was imported into Excel for initial preprocessing. **Key steps included:**

- Handling missing values and removing duplicate entries
- Standardizing date formats and categorical fields (e.g., brand and category names)
- Correcting inconsistent capitalization and trimming whitespace
- Using filters to identify anomalies and invalid entries

These steps ensured a clean, normalized dataset ready for import into SQL for further analysis.

3. SQL – Database Management & Querying -

The cleaned datasets were imported into an SQLite database for structured storage and querying.

Foreign key constraints were checked to ensure referential integrity between tables.

Key SQL Analyses Performed:

- **Store-wise and region-wise sales performance**
- **Product-wise sales and inventory trends**
- **Staff performance evaluation** based on total sales handled
- **Customer order frequency** and purchase patterns
- **Revenue and discount analysis** over time

SQL Views were created to store reusable insights and simplify Power BI integration.

4. Python Integration -

Python (using libraries like pandas, NumPy, and sqlite3) was used to connect to the retail sales database for advanced analysis.

Additional Analytics Included:

- **Monthly revenue trend** and discount impact using SQL queries in Python
- **Customer-level Recency, Frequency, and Monetary (RFM) analysis**
- **Optional segmentation** using KMeans clustering to identify customer groups

The results were exported back to SQLite for visualization in Power BI.

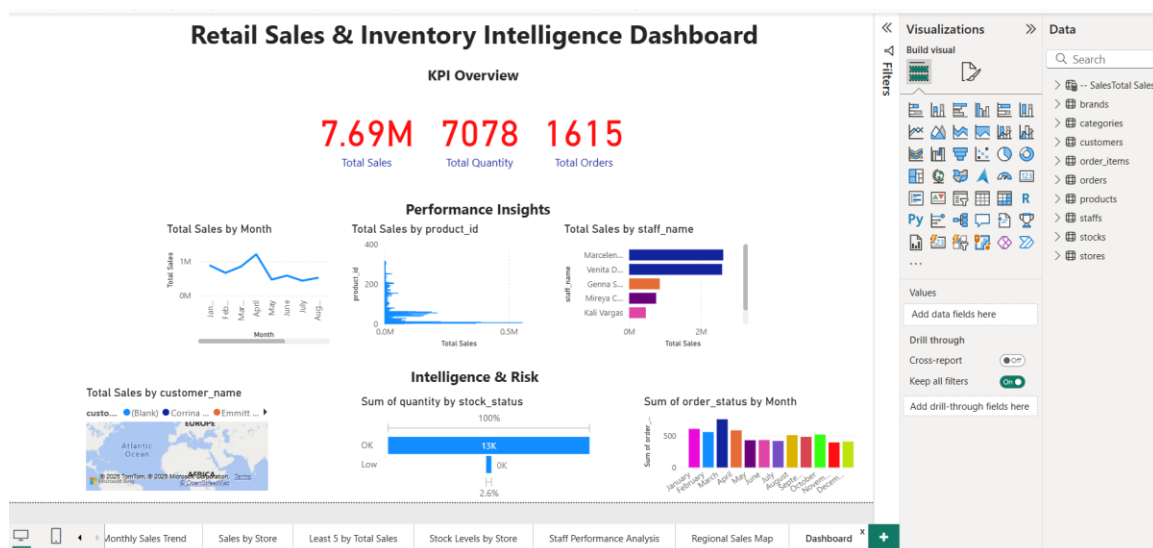
5. Power BI Dashboard -

An interactive Power BI dashboard was created by connecting to the SQLite database views. The dashboard visualized key performance indicators and analytical insights.

Dashboard Visuals:

- **KPIs:** Total Sales, Total Orders, Net Revenue, and Customer Count
- **Bar Charts:** Sales by Product Category, Staff, and Store
- **Maps:** Regional Sales Distribution
- **Line Chart:** Monthly Sales Trends
- **Tables:** Customer and Staff Performance

The dashboard enabled dynamic analysis through filters and drill-down options for store, category, and date.



6. Key Findings -

- The top-selling product categories were **smartphones and accessories**.
- Region-wise analysis showed consistent performance in **urban store locations**.
- Staff with higher **average order values** contributed significantly to revenue.
- Monthly sales trends revealed **seasonal peaks during festivals and holidays**.
- **Discounts increased short-term sales** but slightly reduced overall profit margins.

7. Conclusion -

The Retail Sales & Inventory Intelligence System successfully integrated Excel, SQL, Python, and Power BI to build an end-to-end data analytics pipeline.

The project enabled clear insights into sales performance, inventory management, and customer behavior.

By combining data visualization with SQL and Python-based analytics, the system supports strategic business decisions, operational efficiency, and improved forecasting.